



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

● Experiment No.: 4

1. Title: Use functions types, scope, and closures; apply try-catch; build palindrome checker.

2. Software/Tools Required: VS Code, web browser, HTML, JavaScript.

3. Theory:

1. Regular Function

A regular function is a reusable block of code that performs a specific task. It can accept parameters, return values, and is invoked whenever needed. Functions improve code reusability and organization.

2. Function Declaration

A function declaration defines a function using the function keyword and a function name. It is hoisted, meaning it can be called before its declaration in the code.

3. Function Expression

A function expression stores a function inside a variable. Unlike function declarations, it is not fully hoisted and can only be called after it is defined.

4. Anonymous Function

An anonymous function is a function without a name. It is commonly used as a callback function or assigned to a variable for one-time or temporary operations.

5. Arrow Function

An arrow function is a shorter syntax for writing functions using the => operator. It is concise, easier to read, and automatically inherits the this value from its surrounding scope.

6. Variable Scope

Variable scope determines where a variable can be accessed in a program. JavaScript provides global scope, function scope, and block scope (using let and const).



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4. Output :

File C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac4Palindrome.html

 **Palindrome Checker**

Check Now

✓ **"2442" is a Palindrome**

 Checks Performed : 1

File C:/Users/HP/Desktop/24070521207_SharvariW(JS)/prac4Palindrome.html

 **Palindrome Checker**

Check Now

✗ **"2449" is NOT a Palindrome**

 Checks Performed : 2

5. Result/Conclusion: The JavaScript concepts such as functions, scope, closures, arrow functions, and exception handling were successfully studied and implemented. Their behaviour and usage were verified through simple programs. JavaScript functions and scope mechanisms help in writing clean, reusable, and efficient code. Understanding these concepts improves programming skills and enables the development of reliable and maintainable web app



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Case Study No.: 4

1. **Case Study Title: ATM system verifies if pin is palindrome. if palindrome,mesplay security message . to implement this create a function to reverse the entered pin number and another function to check if it is palindrome. use function declaraction,function expression ,arrow function and demonstrate scope and closure concepts.**

Case Study Program Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>ATM PIN Verification</title>
```

```
<style>
```

```
body {
```

```
font-family: Arial, sans-serif;
```

```
background-color: #f5f0eb; /* Soft beige background */
```

```
color: #4a3c31; /* Deep dark brown text */
```

```
display: flex;
```

```
flex-direction: column;
```

```
align-items: center;
```

```
justify-content: center;
```



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height: 100vh;

margin: 0

}

.card {

background-color: #ffffff;

padding: 30px;

border-radius: 10px;

box-shadow: 0 4px 10px rgba(74, 60, 49, 0.15);

text-align: center;

border-top: 5px solid #6b4c3b; /* Medium warm brown accent */

}

h2 {

color: #5c3a21; /* Rich chocolate brown */

}

label {

display: block;

margin-bottom: 10px;

font-weight: bold;

}



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/* Changed selector from input[type="password"] to input[type="text"] */

input[type="text"] {

padding: 8px;

width: 200px;

border: 1px solid #b8a398;

border-radius: 4px;

margin-bottom: 15px;

}

button {

background-color: #6b4c3b; /* Warm brown button */

color: white;

padding: 9px 18px;

border: none;

border-radius: 4px;

cursor: pointer;

font-weight: bold;

}

button:hover {

background-color: #5c3a21;



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```
}  
  
#result {  
  
    margin-top: 20px;  
  
    font-size: 14px;  
  
}  
  
</style>  
  
</head>  
  
<body>  
  
<div class="card">  
  
    <h2>Sharvari's (1207) PIN Verification</h2>  
  
    <label for="pin">Enter ATM PIN:</label>  
  
    <input type="text" id="pin">  
  
    <button onclick="verifyPIN()">Verify</button>  
  
    <h3 id="result"></h3>  
  
</div>  
  
<script>
```



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// Function Declaration - defines a function with the function keyword and a function name (use when main part of program and you want it available thruout the scope)

```
function reversePIN(pin) {  
  
    return pin.split("").reverse().join("");  
  
}
```

// Function Expression - stores a function inside the variable (use when passing functions as arguments, assigning functions to variables , creating callbacks)

```
const isPalindrome = function(pin) {  
  
    return pin === reversePIN(pin);  
  
};
```

// Arrow Function - shorter way to write a function in js. it was introduced in ES6 and uses arrow syntax

```
const displayMessage = (msg) => {  
  
    document.getElementById("result").innerHTML = msg;  
  
};
```

// Scope and Closure

```
function verifyPIN() {
```



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```
let pin = document.getElementById("pin").value;

function securityMessage() {

    if (isPalindrome(pin)) {

        return " PIN is a Palindrome. Hence, Its a weak password. Please Change it for
max security";

    } else {

        return " PIN is Not a Palindrome. So, security verified!";

    }

}

displayMessage(securityMessage());

}

</script>

</body>

</html>
```



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Output:



Sharvari's (1207) PIN Verification

Enter ATM PIN:

PIN is a Palindrome. Hence, Its a weak password. Please Change it for max security



Sharvari's (1207) PIN Verification

Enter ATM PIN:

PIN is Not a Palindrome. So, security verified!



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3. Result/Conclusion: The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.